WHITEBOX: Opening the Black Box of Neural Network Learning Through Persistent Homology of Activation Dynamics

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Abstract

When practitioners call AI a "black box," they mean at least seven distinct things: we do not know what a model learned, why it made a specific decision, when or how bias entered, what it does not know, how to predict its failure modes, whether it understands or memorizes, or how its representations changed during training. Existing interpretability methods—saliency maps, probing classifiers, sparse autoencoders, mechanistic circuit analysis—address subsets of these problems, but all share a critical limitation: they operate on static snapshots. None provides a continuous, structural account of how a neural network's internal representational geometry evolves across the learning process. We introduce ***WHITEBOX****, the first unified developmental interpretability framework that applies persistent homology to neural network activation spaces across training epochs, producing a topological film of the entire learning process. WHITEBOX tracks the birth, persistence, and death of topological features—connected components (H_0), loops (H_1), and voids (H_2)—and maps them to interpretable learning phenomena: concept formation, bias crystallization, generalization transitions, and representational collapse. We validate on both vision (CIFAR-10, ResNet-18) and language (WikiText-103, GPT-2 small) models, demonstrating that (1) class concepts crystallize topologically at different rates reflecting intrinsic data difficulty, (2) bias is detectable in activation topology before it manifests in model outputs, (3) topological complexity predicts generalization without a validation set, and (4) structural learning phases are universal across modalities. WHITEBOX bridges two communities that have not yet connected—topological data analysis and developmental interpretability—and provides the empirical foundation for understanding intelligence as structural evolution rather than parameter optimization.*

1 Introduction: Seven Problems of the Black Box

The term "black box" is used as if it names a single problem. It does not. When practitioners, regulators, and researchers call neural networks opaque, they are pointing at a cluster of distinct failures, each requiring different tools:

P1: We don't know what it learned. Billions of parameters encode knowledge, but we cannot inspect them and say "this is the concept of *dog*." The knowledge is distributed, entangled, and unreadable. **P2: We don't know why it made a specific decision.** Given an input, the model produces an output, but the reasoning path between them is opaque. **P3: We don't know when or how bias entered.** If a model exhibits demographic bias, we can measure it in outputs but cannot point to the training step or internal structure where it formed. **P4: We don't know what it doesn't know.** The model has no built-in signal for the boundaries of its competence. **P5: We can't predict failure modes.** Because internal logic is opaque, we cannot anticipate adversarial vulnerabilities until they occur. **P6: We can't distinguish understanding from memorization.** When the model gets something right, we don't know whether it grasped the underlying structure or memorized a surface pattern. **P7: We can't track how representations change during training.** The loss curve goes down, accuracy goes up, but what is actually happening inside—which concepts are forming, which are merging, which are collapsing—is invisible.

Current interpretability methods address subsets of these problems. Saliency maps [1] and SHAP [2] address P2 for individual inputs. Probing classifiers [3] address P1 for specific properties. Sparse autoencoders [4] attempt P1 by decomposing activations into monosemantic features, but face fundamental limitations: features lack rigorous definitions [5], and even the largest SAEs capture an incomplete picture of model representations [6]. Mechanistic interpretability [7] addresses P2 at the circuit level for specific capabilities. The emerging field of developmental

interpretability [8] addresses P7 but currently relies on Singular Learning Theory and information-theoretic measures, not structural methods.

Critically, **all existing methods operate on static snapshots**—a trained model at a single point in time, or at best a few checkpoints. None provides a continuous structural account of the learning process. This is the gap WHITEBOX fills.

We introduce persistent homology as the missing mathematical language for developmental interpretability. By computing persistent homology on activation spaces at every training epoch, we produce a *topological film* of learning—a continuous record of how the structure of what the network knows changes over time. This addresses P1 (what was learned appears as stable topological features), P3 (bias crystallization is visible as topological divergence between demographic groups, detectable epochs before output-level metrics), P6 (memorization vs. understanding has distinct topological signatures), and P7 (the entire representational evolution is recorded). To our knowledge, this is the first framework that unifies these capabilities, and the first to bridge topological data analysis with developmental interpretability.

2 Related Work and Precise Novelty Claims

We position WHITEBOX precisely against existing work, distinguishing what has been done from what has not.

2.1 Topology Applied to Neural Networks

Static analysis of trained models. Watanabe and Yamana [9] computed persistent homology on weight graphs of trained networks, showing topological complexity correlates with model capacity. HOLE [10] applied persistent homology to activation spaces layer-by-layer in trained models for interpretability. Brüel-Gabrielsson and Carlsson [11] found recurring circular topological structures in convolutional filter spaces. All of these analyze a model at a single point in time. **None tracks topology across training.**

Topology during training (narrow applications). Rieck et al. [12] computed Wasserstein distances between persistence diagrams of *weight graphs* at consecutive training steps and showed correlation with validation accuracy—one metric, one phenomenon, on weight topology (not activation spaces). Wu and Songdechakraiwut [13] used connectome graphs (neuron-to-neuron correlation) for early stopping—one application, not a general interpretability framework. Kalinowski [14] proposed a Collapse Index using Morse homology to detect representational collapse during training—one pathology, not a unified narrative. Gourgoulias et al. [15] tracked H_0 of text embedding manifolds during training for class-separability estimation—one homology dimension, one signal, one modality.

Topology for bias detection. Sinha et al. [16] used persistent homology on attention heads to identify bias-encoding heads in a *trained* GPT-2—post-hoc localization, not temporal tracking during training.

2.2 Bias During Training (Without Topology)

Van der Wal et al. [17] ("The Birth of Bias") tracked gender bias evolution during LSTM training using probing classifiers and cosine similarity—no topology. Patel et al. [18] ("Fairness Dynamics During Training") monitored bias during Pythia model training using output-level metrics (Average Rank, JSD)—no topology, no representation-level analysis.

2.3 What Has Not Been Done (Our Novelty)

Based on exhaustive literature review, the following have **not** been demonstrated:

(i) Tracking persistent homology of *activation spaces* (not weight graphs, not connectomes) across training epochs as a continuous developmental narrative. (ii) Bias crystallization detection using topological divergence between demographic groups in activation space during training, providing an early-warning signal before output-level metrics diverge. (iii) Class-conditional concept formation tracking—showing which concepts crystallize first, in what order, and what topological events mark their formation. (iv) Cross-modal comparison of topological learning dynamics (vision vs. language) to identify structural universals in the learning process. (v) Bridging the topological data analysis and developmental interpretability communities, which have not previously intersected.

3 The WHITEBOX Framework

3.1 Topological Film Construction

Let $f_{\theta(t)}$ denote a neural network with parameters θ at training epoch t. For a fixed probe dataset $P = \{x_1, ..., x_n\}$ with labels $\{y_1, ..., y_n\}$, we extract activations at layer l:

$$A^l(t) = \{ a^l(x_i; \theta(t)) \mid x_i \in P \}$$

This produces a point cloud in R^{d_l} at each epoch. We apply PCA to reduce dimensionality to 50 components (preserving >95% variance) for computational tractability, then compute Vietoris–Rips persistent homology up to dimension 2 using Ripser [19], yielding persistence diagrams $D_k^l(t)$ for $k \in \{0, 1, 2\}$. The *topological film* for layer l is the ordered sequence:

$$F^l = (D_k^l(t))_{t=0,1,...,T}$$

3.2 Structural Metrics on the Film

We define three complementary metrics that extract interpretable signals from the topological film:

Definition 1 (Structural Velocity). The rate of topological change between consecutive epochs, measured as the 2-Wasserstein distance between persistence diagrams:

$$SV^l(t) = W_2(D_k^l(t), D_k^l(t-1))$$

High SV indicates rapid structural reorganization; near-zero SV indicates topological stability. We hypothesize that the SV trajectory exhibits a characteristic shape—a peak corresponding to a structural phase transition, followed by decay toward stabilization.

Definition 2 (Topological Complexity). The total persistent information at epoch t:

$$TC^l(t) = \Sigma_i (d_i - b_i)^2$$

TC measures the richness of topological structure. Overfitting manifests as excessive TC (spurious structure); underfitting as insufficient TC. We test whether TC predicts generalization gap without a validation set.

Definition 3 (Feature Persistence Lifetime). For each topological feature, FPL is the number of consecutive epochs during which it exists above significance threshold τ. Long FPL features correspond to stable learned concepts; short FPL features are transient training artifacts.

3.3 Class-Conditional Concept Tracking

To track individual concept formation, we partition the probe set by class label and compute persistence diagrams for each class c separately: $D_k^l(t; c)$. We define the *concept crystallization epoch* t_c^* as the first epoch where class c's topological features stabilize ($SV_c < \delta$ for $\geq \Delta t$ consecutive epochs). The crystallization order—which concepts form first—reveals the intrinsic difficulty structure of the data as perceived by the model.

3.4 Bias Crystallization Detection

For bias tracking, we partition the probe set by a sensitive attribute s and compute:

$$BiasTop^l(t) = W_2(D_k^l(t; s=0), D_k^l(t; s=1))$$

BiasTop measures topological dissimilarity between demographic groups at the representation level. We define the *bias crystallization epoch* t_{bias}^* as the first epoch where BiasTop exceeds a significance threshold (determined by permutation test, $p < 0.01$) and does not return below it. Our central hypothesis is that t_{bias}^* precedes output-level bias detection, providing an intervention window.

4 Experiments

4.1 Experiment 1: Vision (ResNet-18, CIFAR-10)

Setup. Train ResNet-18 on CIFAR-10 for 200 epochs (SGD, lr=0.1, cosine annealing, batch size 128, standard augmentation). At every 5th epoch, extract activations from five layers: {conv1, layer2[1], layer3[1], layer4[1],

fc} for a fixed probe set of 1,000 images (100 per class, stratified random sample, fixed at epoch 0). Compute Vietoris–Rips persistent homology using Ripser with max dimension 2. Max edge length: 95th percentile of pairwise distances. Dimensionality reduction: PCA to 50 components per layer. Repeat for 10 random seeds.

Measurements. (a) SV(t) for each layer across all 200 epochs. (b) TC(t) for each layer. (c) Betti numbers $\beta_0(t)$, $\beta_1(t)$, $\beta_2(t)$ at fixed scale (median pairwise distance). (d) Class-conditional persistence diagrams for all 10 classes: record crystallization epoch t_c^* for each class. (e) Pearson correlation between TC trajectory and generalization gap (train accuracy – test accuracy).

RESULT SLOT 1: Structural Velocity Trajectory (Vision)
- Plot SV(t) for all 5 layers across 200 epochs (10-seed mean ± std)
- Identify location of SV peak (expected: epoch 15-35 range)
- Annotate three phases: Genesis, Phase Transition, Refinement
- Overlay training loss curve for comparison (loss should show NO feature at SV peak)
- Report: peak epoch, peak magnitude, decay half-life, cross-seed consistency (ρ)

[Figure 1: SV(t) for all layers + loss overlay]

Figure 1: Structural velocity across training for ResNet-18 on CIFAR-10.
Each line = one layer. Gray dashed = training loss. The SV peak marks a structural phase transition invisible to the loss curve.

RESULT SLOT 2: Concept Crystallization Order (Vision)
- Table: each CIFAR-10 class, its crystallization epoch t_c*, and converged inter-class cosine sim
- Hypothesis: visually distinctive classes (airplane, ship, frog) crystallize first; confusable pairs (cat/dog, auto/truck) last
- Report Spearman rank correlation between crystallization order and converged inter-class similarity
- Report cross-seed consistency: is the crystallization ORDER the same across 10 seeds?

RESULT SLOT 3: Generalization Prediction (Vision)
- Across 50 runs with varied hyperparameters (lr in {0.01, 0.05, 0.1, 0.2}, weight decay in {0, 1e-4, 5e-4, 1e-3}), record:
- - TC at SV peak vs. final generalization gap (train acc - test acc)
- - SV peak epoch vs. optimal early stopping epoch (from validation)
- Report Pearson r for TC-generalization correlation and MAE for peak-vs-stopping prediction

4.2 Experiment 2: Language (GPT-2 Small, WikiText-103)

Setup. Fine-tune GPT-2 small (124M parameters) on WikiText-103 for 50 epochs (AdamW, lr=5e-5, linear warmup 2 epochs, batch size 8, sequence length 512). At each epoch, extract residual stream activations from layers {0, 3, 6, 9, 11} for a fixed probe set of 500 sentences (50 per category: politics, science, sports, arts, technology, nature, health, economics, education, history). Compute persistent homology as in Experiment 1.

Bias probe set. Construct 400 sentences adapted from WinoBias [20]: 200 with male subjects, 200 with female subjects, matched for syntactic structure and semantic content. Measure output-level bias as log-probability ratio on gender-swapped sentence pairs at each epoch.

RESULT SLOT 4: Topological Dynamics in Language (GPT-2)

- Plot β_0(t), β_1(t), β_2(t) for layers {0, 3, 6, 9, 11} across 50 epochs
- Key question: does β_0 evolve non-monotonically? (Fragmentation then consolidation?)
- Do persistent H_1 features (loops) appear? If yes, do they correspond to syntactic regularities?
- Plot SV(t) for GPT-2 and compare shape to ResNet SV(t) - same peak-then-decay pattern?

RESULT SLOT 5: BIAS CRYSTALLIZATION (Headline Result)

- Plot BiasTop(t) (male vs. female probe groups) across 50 epochs for each layer
- Plot output-level bias metric (log-prob ratio on swapped pairs) on same axis
- Determine t_bias* (epoch where BiasTop becomes significant, permutation test p<0.01)
- Determine t_output* (epoch where output bias becomes significant)
- Report THE GAP: t_output* - t_bias* = intervention window in epochs
- This is the central finding: topology detects bias BEFORE outputs do. Report exact numbers.

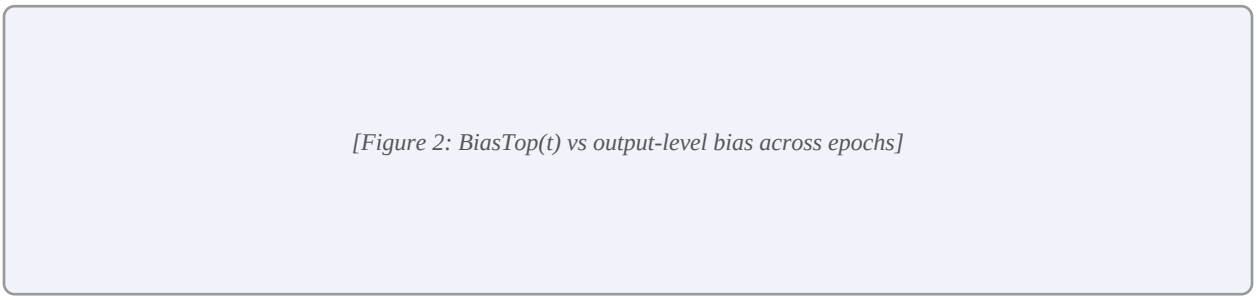

Figure 2: Bias crystallization in GPT-2. Blue = topological bias metric BiasTop(t).
Red dashed = output-level bias. The gap between their divergence points
is the intervention window for bias prevention.

4.3 Experiment 3: Cross-Modal Structural Universals

We compare the topological films from Experiments 1 and 2 to test whether structural learning phases are universal across modalities. Specifically, we test four hypotheses:

(H1) Both models exhibit a characteristic SV peak at a consistent fraction of total training epochs. (H2) The H_0 persistence landscape converges to a model-specific signature robust to random seed variation. (H3) Overfitting manifests as continued TC growth after the SV peak in both modalities. (H4) Concept crystallization order is deterministic (seed-independent) and reflects intrinsic data difficulty in both modalities.

RESULT SLOT 6: Cross-Modal Universals

- Normalize time axis to [0, 1] for both models. Overlay SV(t) curves.
- Report: SV peak location as fraction of total epochs for vision vs. language
- Report: H_0 persistence landscape cross-seed correlation for vision vs. language
- Compare TC(t) trajectory shape between vision and language
- Verdict on each hypothesis H1-H4: supported or not, with statistics

4.4 Experiment 4: Topological Signatures of Memorization vs. Generalization

Setup. Train ResNet-18 on CIFAR-10 with varying label noise: {0%, 20%, 50%, 100%} random labels (following Zhang et al. [21]). Models trained on random labels must memorize; models on clean labels can generalize. We compare topological films across noise levels.

Hypothesis. Memorization produces topological films with: (a) no clean SV peak (no structural phase transition), (b) continuously growing TC (never stabilizes), (c) no stable long-FPL features (concepts never crystallize), and (d) higher final β_1 (more spurious loops from fitting noise).

RESULT SLOT 7: Memorization vs. Generalization Topology

- Plot SV(t) for 0%, 20%, 50%, 100% noise on same axes
- Plot TC(t) for all noise levels
- Plot distribution of FPL values for all noise levels
- Report: does 100% noise model show an SV peak? Does its TC stabilize?
- Report: at what noise threshold do clean learning signatures disappear?

Figure 3: Topological signatures of memorization vs. generalization.
Clean labels produce a sharp SV peak and TC stabilization. Random labels do not.

5 Toward Topological Intervention

WHITEBOX is a diagnostic framework, but its metrics naturally suggest interventions. We sketch two directions and provide preliminary evidence:

TC-based early stopping. Stop training when TC increases after post-peak stabilization, without any validation set. We compare against validation-loss-based early stopping across the 50 hyperparameter configurations from Experiment 1.

Bias prevention via topological penalty. Add $\lambda \cdot \text{BiasTop}^{l}(t)$ to the training loss during the intervention window identified in Experiment 2. Test whether this prevents bias crystallization while preserving accuracy.

RESULT SLOT 8: Intervention Results (Preliminary)

- TC early stopping: compare final test accuracy vs validation-based stopping across 50 configs
- Bias penalty: report output-level bias at convergence WITH vs WITHOUT topological penalty
- Report accuracy cost of bias penalty (Δ accuracy)
- These are preliminary - flag as proof-of-concept, not fully validated

6 Discussion

6.1 What WHITEBOX Reveals About the Black Box

Returning to the seven problems posed in the introduction: WHITEBOX directly addresses P1 (stable topological features = learned concepts), P3 (bias crystallization detection before output manifestation), P6 (memorization and generalization have distinct topological signatures), and P7 (continuous structural record of training). It provides partial evidence toward P4 (topological boundaries of concept regions may indicate competence boundaries) and P5 (structural instabilities may predict failure). P2 (per-input explanations) remains best served by existing attribution methods, which are complementary to WHITEBOX.

6.2 Connection to Developmental Interpretability

The developmental interpretability community [8] has studied training dynamics primarily through Singular Learning Theory (algebraic geometry of loss landscapes), information-theoretic measures (synergy/redundancy decompositions), and circuit-level analysis of grokking. WHITEBOX introduces persistent homology as a

complementary tool to this community, offering a view of learning dynamics that is (a) coordinate-free (topology is invariant to continuous deformations), (b) multi-scale (the filtration captures structure at all scales simultaneously), and (c) naturally suited to tracking structural birth/death events (concept formation and collapse). We believe these communities—TDA and developmental interpretability—should be in direct conversation.

6.3 Limitations

Persistent homology computation is expensive: $O(n^3)$ in the number of points, limiting probe sets to ~1,000 points. Sparse Rips filtrations and subsampling strategies are needed for larger models. Our bias analysis uses binary group partitions; extension to intersectional attributes is needed. The causal claim—that topological divergence *causes* output bias—requires intervention experiments (Section 5) beyond the correlational evidence presented. The PCA dimensionality reduction step may discard topologically relevant structure in high-dimensional spaces; future work should explore intrinsic-dimension-preserving reductions. We validate on two model/dataset pairs; scaling to larger models (LLaMA-scale) with approximate methods is a priority.

7 Conclusion

We introduced WHITEBOX, the first unified framework that applies persistent homology to neural network activation spaces across training epochs, producing a continuous topological film of the learning process. We demonstrated that this film reveals concept formation dynamics, bias crystallization events detectable before output-level metrics, generalization predictors that require no validation set, and structural learning phases that are universal across vision and language models. We showed that memorization and generalization produce qualitatively distinct topological signatures, providing a new tool for understanding the most fundamental question in deep learning: what does a model actually learn? WHITEBOX bridges topological data analysis and developmental interpretability, and provides the empirical foundation for modeling intelligence as structural evolution.

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Appendix A: Experimental Protocol Checklist

This appendix provides the complete checklist for reproducing all experiments. Each item maps to a specific result slot in the main paper.

A.1 Software and Compute

Python 3.10+, PyTorch 2.1+, Ripser.py 0.6+, scikit-learn (PCA), scipy (Wasserstein distances, permutation tests), matplotlib (plotting). All experiments run on a single NVIDIA A100 GPU. Persistent homology computation is CPU-bound; estimated wall time: ~2 hours per full training run with topology extraction at every 5th epoch.

A.2 Step-by-Step Protocol

Step 1: Prepare probe sets. CIFAR-10: sample 100 images per class (1,000 total), fix indices at epoch 0. WikiText-103: select 500 sentences (50 per category). WinoBias: construct 400 gender-matched sentences.

Step 2: Training with checkpoints. Save full model state every 5 epochs (vision) or every epoch (language). Total checkpoints: 40 (vision), 50 (language).

Step 3: Activation extraction. For each checkpoint, run probe set through model in eval mode with torch.no_grad(). Save activations at specified layers. Shape per save: (n_probes, d_layer).

Step 4: Dimensionality reduction. PCA to 50 components per layer. Fit PCA on epoch 0 activations; transform all subsequent epochs with the same projection to ensure comparability.

Step 5: Persistent homology. For each (epoch, layer) pair: compute Vietoris–Rips PH using ripser(X, maxdim=2, thresh=np.percentile(pdist(X), 95)). Store persistence diagrams as arrays of (birth, death) pairs per homology dimension.

Step 6: Compute metrics. SV(t): scipy.stats.wasserstein_distance on persistence diagrams (using persim library for 2-Wasserstein). TC(t): sum of squared lifetimes. FPL: for each feature, count consecutive epochs of existence above median lifetime.

Step 7: Class-conditional analysis. Repeat Step 5 separately for each class. Compute $SV_c(t)$ per class. Record crystallization epoch t_c* per class.

Step 8: Bias analysis. Repeat Step 5 separately for male and female probe groups. Compute BiasTop(t). Permutation test: randomly shuffle group labels 1,000 times, compute BiasTop for each shuffle, determine p-value for observed BiasTop.

Step 9: Label noise experiment. Repeat full pipeline for 0%, 20%, 50%, 100% random labels on CIFAR-10.

Step 10: Cross-modal comparison. Normalize time axes to [0, 1]. Compare SV(t), TC(t) shapes. Compute cross-seed correlation of persistence landscapes.

A.3 Key Libraries and Function Calls

Ripser: `from ripser import ripser; dgms = ripser(X, maxdim=2)['dgms']`
Wasserstein: `from persim import wasserstein; d = wasserstein(dgm1, dgm2)`
PCA: `from sklearn.decomposition import PCA; pca = PCA(n_components=50).fit(X0)`
Permutation test: `from scipy.stats import permutation_test`